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classmate

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* Least squares estimate for nested models

$$Y = XB + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I)$$

$n \times 1$ $n \times p$ $p \times 1$ $n \times 1$

Full model: $Y = \beta_0 + \beta_1 X^{(1)} + \dots + \beta_{p-1} X^{(p-1)} + \varepsilon$

Submodel: $Y = \beta_0 + \beta_1 X^{(1)} + \dots + \beta_{q-1} X^{(q-1)} + \varepsilon \quad (q < p)$

Design matrix for full model: $X = [1: X^{(1)} \dots X^{(p-1)}]$

" " " sub " : $Z = [1: X^{(1)} \dots X^{(q-1)}]$

$n \times p$ $n \times q$

Z consists of a subset of columns of X

We may write: $X = [Z: W]$

$n \times p$ $n \times q$ $n \times (p-q)$

~~$Y = X\beta + \varepsilon$~~ is full model -- (i)

$Y = Z\beta + \varepsilon$ is sub model -- (ii)

• If $C(Z) \subset C(X)$, then (ii) is "nested" within (i).

→ spc. $X = [Z: W]$ and $P(X) = P$, so that $P(Z) = q$ & $P(W) = r$.

$n \times p$ $n \times q$ $n \times r$ $(p = q + r)$

Goal: We want to express the least squares estimate of γ in model (ii) in way that relates it to the least squares estimate of β in model (i).

Lemma: $P_X = P_Z + P_{W|Z}$ where $P_{W|Z}$ is the orthogonal matrix onto $C((I - P_Z)W)$ & P_X, P_Z are ~~res~~ respective orthogonal projectors onto their column spaces.

Proof: Let $u \in \mathcal{C}(X)$. We want to show that
 $u = P_Z u + P_{W|Z} u$

If $u \in \mathcal{C}(X)$, then $u = Zv_1 + Wv_2$ for some $v_1 \in \mathbb{R}^n$
 $v_2 \in \mathbb{R}^r$

$$\Rightarrow (I - P_Z)u = (I - P_Z)Wv_2 \quad ((I - P_Z)Zv_1 = 0)$$

Also, $u = P_Z u + (I - P_Z)u = P_Z u + (I - P_Z)Wv_2$

Claim: $(I - P_Z)Wv_2 = P_{W|Z} u$

$$\begin{aligned} P_{W|Z} u &= (I - P_Z)W(W^T(I - P_Z)W)^{-1}W^T(I - P_Z)u \\ &= (I - P_Z)W(W^T(I - P_Z)W)^{-1}W^T(I - P_Z)Wv_2 \\ &= (I - P_Z)Wv_2 \end{aligned}$$

$\therefore u = P_Z u + P_{W|Z} u$ □

→ Recall: $Y = X\beta + \varepsilon$, $Y = Z\gamma + \varepsilon$

$$\therefore X = [Z; W], \quad \beta = \begin{bmatrix} \gamma \\ \delta \end{bmatrix}$$

$$\therefore Y = X\beta + \varepsilon = Z\gamma + W\delta + \varepsilon$$

- ~~Obs~~ Observation: If $Z^T W = 0$ then,
 LSE of β is $\hat{\beta} = \begin{pmatrix} \hat{\gamma} \\ \hat{\delta} \end{pmatrix}$ where $\hat{\gamma}$ is the

LSE of γ from (ii) and $\hat{\delta}$ is the LSE of δ
 from $Y = W\delta + \varepsilon$ -- (iii). This is because the LS
 criterion for model (i) is:

$$\begin{aligned} \|Y - X\beta\|^2 &= \|Y - Z\gamma - W\delta\|^2 \\ &= \|(I - P_X)Y + P_X Y - (Z\gamma + W\delta)\|^2 \end{aligned}$$

$$= \|(I - P_X)Y\|^2 + \|(P_Z Y + P_W Y) - (Z\gamma + W\delta)\|^2$$

$$= \|(I - P_X)Y\|^2 + \|P_Z Y - Z\gamma\|^2 + \|P_W Y - W\delta\|^2$$

$$\therefore Z^T W = 0, P_X = P_Z + P_W \text{ (by lemma)}$$

Correspondingly, LSE of γ : minimizes the criterion

$$\|Y - Z\gamma\|^2 = \|(I - P_Z)Y\|^2 + \|P_Z Y - Z\gamma\|^2$$

Propⁿ: Under the assumption of the lemma, the LSE of $\beta = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$ from the full model (i) can be expressed as

$$\hat{\beta} = \begin{pmatrix} \hat{\gamma} \\ \hat{\delta} \end{pmatrix} \text{ where, } \left. \begin{aligned} \hat{\gamma} &= (Z^T(I - P_W)Z)^{-1}Z^T(I - P_W)Y \\ \hat{\delta} &= (W^T(I - P_Z)W)^{-1}W^T(I - P_Z)Y \end{aligned} \right\} \text{---(iv)}$$

Notice that, in general $\hat{\gamma} \neq \tilde{\gamma}$ = LSE of γ from model (ii)
& $\hat{\delta} \neq \tilde{\delta}$ = " " " " " (iii)

$$\tilde{\gamma} = (Z^T Z)^{-1}Z^T Y \text{ \& } \tilde{\delta} = (W^T W)^{-1}W^T Y$$

Remark: Consider the transformed data models:

$$(I - P_W)Y = (I - P_W)Z\gamma + \varepsilon' \text{ --- (ii)'}$$

$$(I - P_Z)Y = (I - P_Z)W\delta + \varepsilon'' \text{ --- (iii)'}$$

$$\begin{aligned} \text{LSE of } \gamma \text{ from (ii)' is } (Z^T(I - P_W)^T(I - P_W)Z)^{-1}Z^T(I - P_W)^T(I - P_W)Y \\ = (Z^T(I - P_W)Z)^{-1}Z^T(I - P_W)Y = \hat{\gamma} \end{aligned}$$

similarly for $\hat{\delta}$.

→ We get $\hat{\gamma}, \hat{\delta}$ by removing the effect of the other variables (both predictor and response) and working with these new models.

Proof of Propⁿ $\hat{\beta}$ is the solution of the normal equation.

$$X^T X \hat{\beta} = X^T Y$$

We know: $X \hat{\beta} = P_X Y = P_Z Y + P_{W|Z} Y$
 $\parallel$
 $Z \hat{\gamma} + W \hat{\delta}$

Multiply both sides by $(I - P_Z)$:

$$\Rightarrow (I - P_Z) W \hat{\delta} = (I - P_Z) P_{W|Z} Y$$

$$= (I - P_Z) (I - P_Z) W (W^T (I - P_Z) W)^{-1} W^T (I - P_Z) Y$$

Multiply by W^T :

$$\Rightarrow W^T (I - P_Z) W \hat{\delta} = W^T (I - P_Z) Y$$

$$\Rightarrow \hat{\delta} = (W^T (I - P_Z) W)^{-1} W^T (I - P_Z) Y$$

Similarly for $\hat{\gamma}$. 

→ Implications:

- (i) R^2 statistic: When there is 1 predictor variable,
 $R^2 = (\text{corr}(X, Y))^2$

More generally, $R^2 =$ proportion of variability in the data (response) explained by the predictor predictor variables in the linear model.

$$Y = \beta_0 + \beta_1 X^{(1)} + \dots + \beta_{p-1} X^{(p-1)} + \epsilon$$

$$\text{Total variability in } Y = \sum (Y_i - \bar{Y})^2 = \sum Y_i^2 - n \bar{Y}^2$$

$$= \|Y\|^2 - \frac{1}{n} (1^T Y)^2$$

$$= Y^T Y - Y^T \left(\frac{1}{n} \mathbf{1} \mathbf{1}^T \right) Y$$

$$\hookrightarrow P_Z$$

$$= Y^T(I - P_2)Y = SST$$

$$\begin{aligned} \text{For } p=2, R^2 &= \frac{(\text{Cov}(X, Y))^2}{\text{Var}(X) \cdot \text{Var}(Y)} \\ &= \frac{(\sum (X_i - \bar{X})(Y_i - \bar{Y}))^2}{\sum (X_i - \bar{X})^2 \cdot \sum (Y_i - \bar{Y})^2} \\ \Rightarrow R^2 &= \frac{\hat{\beta}_1^2 \sum (X_i - \bar{X})^2}{\sum (X_i - \bar{Y})^2} \rightarrow \begin{matrix} SSR \\ SST \end{matrix} \end{aligned}$$

$$\text{Anova: } SST = SSR + SSE$$

$$\begin{aligned} SSE &= \sum (Y_i - \hat{Y}_i)^2 \\ &= \|Y - \hat{Y}\|^2 = \|Y - P_X Y\|^2 \\ &= Y^T(I - P_X)Y \end{aligned}$$

$$\therefore SSR = Y^T(P_X - P_2)Y$$

Comparison of R^2 values for full model and sub model:

$$Y = \beta_0 + \beta_1 X^{(1)} + \dots + \beta_{p-1} X^{(p-1)} + \epsilon = X\beta + \epsilon$$

$$Y = \beta_0 + \beta_1 X^{(1)} + \dots + \beta_{q-1} X^{(q-1)} + \epsilon = Z\gamma + \epsilon$$

$$R_{full}^2 = \frac{Y^T(P_X - P_2)Y}{Y^T(I - P_2)Y} ; R_{sub}^2 = \frac{Y^T(P_Z - P_2)Y}{Y^T(I - P_2)Y}$$

$$\text{Difference of numerators} := Y^T(P_X - P_Z)Y = Y^T \tilde{P} Y$$

projection matrix

$$\text{If } X = [Z : W], \text{ then } \tilde{P} = P_{(I - P_Z)W} = P_{W|Z}$$

$\hookrightarrow$ projⁿ matrix, hence n.n.d.

$$\therefore Y^T \tilde{P} Y \geq 0$$

$$\therefore R_{full}^2 \geq R_{sub}^2 //$$

If we add extra predictor variables, values of R^2 never decreases